

Tute (Week 2)

Cardinalities

Question 1

Find the cardinalities of the following sets

1. $\left\{ \frac{1}{n} : n \in \mathbb{N} \text{ and } n \in [1, 4] \right\}$
2. $\left\{ n^2 - n : n \in \mathbb{N} \text{ and } n \in [0, 4] \right\}$
3. $\left\{ \frac{1}{n^2} : n \in \mathbb{N}_{>0} \text{ and } 2|n \text{ and } n < 11 \right\}$
4. $\{2 + (-1)^n : n \in \mathbb{N}\}$

1. This is just $\left\{ \frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \right\}$, so the cardinality is 4.
2. This is just $\{0, 0, 2, 6, 12\}$, so the cardinality is 4.
3. This is just $\left\{ \frac{1}{4}, \frac{1}{16}, \frac{1}{36}, \frac{1}{64}, \frac{1}{100} \right\}$ so the cardinality is 5.
4. This is just $\{3, 1, 3, 1, \dots\}$ so the cardinality is 2.

Question 2

If $|A| = m$, $|B| = n$ and $|A \cap B| = k$, what is:

1. $|A \cup B|$
2. $|A \setminus B|$
3. $|A \oplus B|$

1. $|A \cup B| = |A| + |B| - |A \cap B| = m + n - k$
2. $|A \setminus B| = |A| - |A \cap B| = m - k$
3. $|A \oplus B| = |A \cup B| - |A \cap B| = m + n - 2k$

Question 3

How many elements in the following sets:

1. $A = \{w \in \Sigma^* : \text{length}(w) \leq 3\}$ where $\Sigma = \{a, b, c\}$
2. $B = \{w \in \Sigma'^* : \text{length}(w) \in [2, 4]\}$ where $\Sigma' = \{a, b\}$
3. $A \oplus B$

4. $AB := \{wv : w \in A \text{ and } v \in B\}$ *Hint: it's less than $40 \times 28 = 1120$*

The general approach here is to tally up the words of each possible length separately.

1. A has $3^0 + 3^1 + 3^2 + 3^3 = 1 + 3 + 9 + 27 = 40$ elements.

2. B has $4 + 8 + 16 = 28$ elements

3. Since B has no words of length 0 and 1, we only need to tally up the words from A. For lengths 2 and 3 we have that the B-words are a subset of the A-words, so to get the words that only occur in A we subtract the number of A-words from the number of B-words. Finally, since A has no words of length 4, we only need to count the B-words. Thus:

$$\begin{aligned} \text{card}(A \oplus B) &= \\ 1 + 3 + (9 - 4) + (27 - 8) + 16 &= \\ 44 \end{aligned}$$

4. It is certainly fewer than $40 \times 28 = 1120$ because $wv = w'v'$ when, e.g. $w = \lambda, v = aaa, w' = a, v' = aa$. We need to count all the words over $\{a, b, c\}$, with length between 2 and 7, where c's can only occur up to the position $\min\{3, \text{length}(w) - 2\}$

i. Length 2 (no c's): $2 \times 2 = 4$ words

ii. Length 3 (c only possible in first position): $3 \times 2 \times 2 = 12$

iii. Length 4 (c only possible in first two positions): $3 \times 3 \times 2 \times 2 = 36$

iv. Length 5 (c only possible in first three positions): $3^3 \times 2^2 = 108$

v. Length 6 (c only possible in first three positions): $3^3 \times 2^3 = 216$

vi. Length 7 (c only possible in first three positions): $3^3 \times 2^4 = 432$

Total: 808.

Set Reasoning

Question 1

When is $(A \setminus B) \setminus C = A \setminus (B \setminus C)$?

As can be checked with a Venn diagram:

$$A \setminus (B \setminus C) = ((A \setminus B) \setminus C) \cup (A \cap C)$$

so when $A \cap C = \emptyset$ we have equality.

Question 2

Show, using the Laws of Set Operations (and Idempotence) the following equalities hold for all sets A and B (subsets of $\square$):

1. Annihilation: $A \cap \emptyset = \emptyset$ and $A \cup \square = \square$
2. Double complementation: $(A^c)^c = A$ *Hint*: First show that if $A \cap B = \emptyset$ and $A \cup B = \square$ then $B = A^c$
3. De Morgan's Laws: $(A \cap B)^c = A^c \cup B^c$ and $(A \cup B)^c = A^c \cap B^c$ *Hint*: the lemma we used to prove double complementation is useful here too. *Hint*: by the principle of duality it suffices to prove one of the laws.

Note: using the Principle of Duality, we only need to show one equality for each part.

Annihilation (Part 1):

$$\begin{aligned}
 A \cap \emptyset &= A \cap (A \cap A^c) && \text{(Complementation)} \\
 &= (A \cap A) \cap A^c && \text{(Associativity)} \\
 &= A \cap A^c && \text{(Idempotence)} \\
 &= \emptyset && \text{(Complementation)}
 \end{aligned}$$

Double Complementation (Part 2):

Following the hint, if $A \cap B = \emptyset$ and $A \cup B = \square$:

$$\begin{aligned}
 A^c &= A^c \cap \square && \text{(Identity)} \\
 &= A^c \cap (A \cup B) && \text{(Assumption)} \\
 &= (A^c \cap A) \cup (A^c \cap B) && \text{(Distributivity)} \\
 &= (A \cap A^c) \cup (A^c \cap B) && \text{(Commutativity)*} \\
 &= \emptyset \cup (A^c \cap B) && \text{(Complementation)} \\
 &= (A^c \cap B) \cup \emptyset && \text{(Commutativity)*} \\
 &= A^c \cap B && \text{(Identity)} \\
 &= B \cap A^c && \text{(Commutativity)} \\
 &= (B \cap A^c) \cup \emptyset && \text{(Identity)} \\
 &= (B \cap A^c) \cup (B \cap A) && \text{(Assumption)} \\
 &= B \cap (A^c \cup A) && \text{(Distributivity)} \\
 &= B \cap (A \cup A^c) && \text{(Commutativity)*} \\
 &= B \cap \square && \text{(Complementation)} \\
 &= B && \text{(Identity)}
 \end{aligned}$$

Now we have, using Commutativity and Complementation:

$$\begin{aligned}
 A^c \cap A &= A \cap A^c = \emptyset \\
 A^c \cup A &= A \cup A^c = \square
 \end{aligned}$$

So $A = (A^c)^c$.

De Morgan's Laws (Part 3):

We have:

$$\begin{aligned}
(A \cap B) \cap (A^c \cup B^c) &= A \cap (B \cap (A^c \cup B^c)) && \text{(Associativity)} \\
&= A \cap ((B \cap A^c) \cup (B \cap B^c)) && \text{(Distributivity)} \\
&= A \cap ((B \cap A^c) \cup \emptyset) && \text{(Complementation)} \\
&= A \cap (B \cap A^c) && \text{(Identity)} \\
&= (A \cap B) \cap A^c && \text{(Associativity)} \\
&= (B \cap A) \cap A^c && \text{(Complementation)} \\
&= B \cap (A \cap A^c) && \text{(Associativity)} \\
&= B \cap \emptyset && \text{(Complementation)} \\
&= \emptyset && \text{(Annihilation: see (a))}
\end{aligned}$$

By the Principle of Duality we also have

$$(X \cup Y) \cup (X^c \cap Y^c) = \square$$

So,

$$\begin{aligned}
(A \cap B) \cup (A^c \cup B^c) &= (A^c \cup B^c) \cup (A \cap B) && \text{(Commutativity)} \\
&= (A^c \cup B^c) \cup ((A^c)^c \cap (B^c)^c) && \text{(Double complement)} \\
&= \square && \text{(from above)}
\end{aligned}$$

It follows from the previous part (Uniqueness of complement) that $(A \cap B)^c = (A^c \cup B^c)$.

Question 3

Prove, using the Laws of Set Operations (and derived laws), that:

$$(A \cap B^c) \cup (B \cap A^c) = (A \cup B) \cap (A \cap B)^c$$

$$\begin{aligned}
&(A \cap B^c) \cup (B \cap A^c) \\
&= ((A \cap B^c) \cup B) \cap ((A \cap B^c) \cup A^c) && \text{(Distributivity)} \\
&= (B \cup (A \cap B^c)) \cap (A^c \cup (A \cap B^c)) && \text{(Commutativity)} \\
&= ((B \cup A) \cap (B \cup B^c)) \cap ((A^c \cup A) \cap (A^c \cup B^c)) && \text{(Distributivity)} \\
&= ((B \cup A) \cap (B \cup B^c)) \cap ((A^c \cup (A^c)^c) \cap (A^c \cup B^c)) && \text{(Dbl. compl.)} \\
&= ((B \cup A) \cap \square) \cap (\square \cap (A^c \cup B^c)) && \text{(Complement)} \\
&= ((A \cup B) \cap \square) \cap ((A^c \cup B^c) \cap \square) && \text{(Commutativity)} \\
&= (A \cup B) \cap (A^c \cup B^c) && \text{(Identity)} \\
&= (A \cup B) \cap (A \cap B)^c && \text{(De Morgan's law)}
\end{aligned}$$

Relations and Orders

Question 1

Let Σ be a finite alphabet.

We define a relation $R \subseteq \Sigma^* \times \Sigma^*$ as follows:

$$(w, v) \in R \quad \text{if, and only if,} \quad \exists x \in \Sigma^*. (wx = v)$$

Show that R is a partial order.

To show that R is a partial order, we need to show that it is reflexive, antisymmetric, and transitive.

1. **Reflexive:** For all $w \in \Sigma^*$ we observe that $w = w\lambda$. Hence, $(w, w) \in R$, and so R is reflexive.
2. **Antisymmetric:** Suppose $(w, v) \in R$ and $(v, w) \in R$. Then there exists $x, y \in \Sigma^*$ such that $v = wx$ and $w = vy$. So $w = wxy$, and so $\text{length}(xy) = 0$ and $xy = \lambda$. Therefore $x = y = \lambda$ so $w = v\lambda = v$. So R is antisymmetric.
3. **Transitive:** Suppose $(w, v), (v, x) \in R$. Then there exists y, z such that $wy = v$ and $vz = x$. But then $w(yz) = (wy)z = vz = x$ so $(w, x) \in R$. Hence R is transitive.